

Conservation Genetics Homework #1

Name: _____

- 1) Imagine a mutation produces a third allele at frequency r that also affects color morphs in Ferruginous Hawks. Expand your 'population' Punnett Square with an extra row and column for r and determine all possible crosses.

a. What are the expected genotype frequencies after one generation?

- b. Given allele frequencies $p = 0.6$, $q = 0.2$, and $r = 0.2$, how many chicks will be homozygous if 200 are born this year?

- 2) A graduate student in the Department of Ecology genotypes 400 Ferruginous Hawks in Oregon ("Pop 1") and 400 Ferruginous Hawks east of the Continental Divide in Montana ("Pop 2"). In Pop 1 they find 324 birds are homozygous recessive, 72 birds are heterozygous, and 4 birds are homozygous dominant. In Pop 2 they find 196 birds are homozygous recessive, 168 birds are heterozygous, and 36 birds are homozygous dominant. Determine the total number of A and a alleles in each subpopulation, as well as their relative frequencies. Next, pool the allele counts for Pop 1 and Pop 2 and do the same for the "global" population of hawks in the study (i.e., treat all hawks as belonging to the same population).

a. What are the expected genotype frequencies of Pop 1 under HWE?

$$f(AA) = \underline{\hspace{2cm}} \qquad f(Aa) = \underline{\hspace{2cm}} \qquad f(aa) = \underline{\hspace{2cm}}$$

b. What are the expected genotype frequencies of Pop 2 under HWE?

$$f(AA) = \underline{\hspace{2cm}} \qquad f(Aa) = \underline{\hspace{2cm}} \qquad f(aa) = \underline{\hspace{2cm}}$$

c. What are the expected genotype frequencies of the global population under HWE?

$$f(AA) = \underline{\hspace{2cm}} \qquad f(Aa) = \underline{\hspace{2cm}} \qquad f(aa) = \underline{\hspace{2cm}}$$

- d. Is Pop 1 currently under HWE?* (YES / NO)
Is Pop 2 currently under HWE?* (YES / NO)
Is the global population currently in HWE?* (YES / NO)

* *don't worry about performing a χ^2 test!*